



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**FIRST YEAR FIRST SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE, BACHELOR OF
EDUCATION SCIENCE WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

SZL 101: LOWER INVERTEBRATES

Date: 20th February, 2018

Time: 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any TWO questions in SECTION B.
- Use illustrations where appropriate.



SECTION A (60 Marks):

INSTRUCTIONS: ANSWER ALL QUESTIONS IN THIS SECTION

Q1. Briefly describe the different ways by which unicellular organisms take in food and other substances.

Q2. Give a brief description of the composition and function of the Kinet system in ciliates.

Q3. Describe the composition and main functions of the epithelia in invertebrates.

Q4. Describe the main distinguishing features of the Phylum Dinophyta.

Q5. Using a specific example, describe the life cycle of the sporozoid protozoa.

Q6. Describe the structure and formation of the THREE photoreceptors found in invertebrates and name ONE species each in which they are found.

Q7. Using specific examples, briefly explain the significance of radial symmetry in lower invertebrates.

Q8. Name and state the functions of FIVE principal types of cells found in the epidermis of Hydrozoans.

Q9. State the functions of the following in invertebrates:

- | | |
|----|---------------|
| a) | Nematodesmata |
| b) | Rod Organ |
| c) | Peritonium |
| d) | Rhabdites |
| e) | Vitellarium |
| f) | Dactylozooids |

Q10. Distinguish between filtration and secretion kidneys in invertebrates.

SECTION B:

INSTRUCTIONS: Answer any TWO questions (20 Marks each question)

Q11. Give a detailed account of the basic structure and physiology of Tapeworms (Class Cestoidea).

Q12. Discuss the importance of cephalisation and compartmentation in bilateral animals.

Q13. Describe the criteria used to classify the amoebic protozoa.

Q14. Discuss 'polymorphism' in Hydrozoans.



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MAIN CAMPUS

SZL 103: INTRODUCTORY GENETICS AND EVOLUTION

Date: 14th February, 2018

Time: 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any TWO question in SECTION B.
- Illustrate your answers with relevant examples and/or well labeled diagrams where appropriate.



Section A. (40 Marks)

Answer all questions in this section. All questions carry equal marks (5 Marks).

- 1) Define the terms homozygous, heterozygous, dominant, recessive and locus.
- 2) Give any **five** reasons as to why Gregor Mendel succeeded in his experiments on inheritance in plants.
- 3) State any **five** tenets of the modern evolutionary synthesis.
- 4) State any **six** limitations of using fossils to show evolutionary relationships.
- 5) Outline Darwin's concept of evolution through natural selection.
- 6) State any **six** conclusions that Mendel made from his experiments.
- 7) Briefly explain test cross and its significance in identifying genotypes.
- 8) Briefly state any **three** methods that can be used to make transgenic animals.

SECTION B. 30 Marks

Answer any **two** questions. Each question carries 15 Marks

- 9) Discuss the main forms of pre-zygotic and post-zygotic isolation mechanisms.
- 10) Discuss the importance of Hardy-Weinberg concept in the understanding of natural selection and evolution.
- 11) Using relevant examples, discuss adaptive radiation and how it results into the development of new species.
- 12) Discuss the theory that the genus *Homo* evolved in Africa.



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**SECOND YEAR FIRST SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE WITH INFORMATION
TECHNOLOGY**

MAIN CAMPUS

SZL 201: CELL BIOLOGY

Date: 14th February, 2018

Time: 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any TWO question in SECTION B.
- Illustrate your answers with relevant examples and/or well labeled diagrams where appropriate.



SECTION A

Answer all questions in this section (60 marks). All questions carry equal marks

1. Draw a labelled diagram of a typical bacterium and outline 2 reasons that qualify it to be a cell.
2. State and briefly explain three factors that limit cell size.
3. Briefly describe the role of microtubules in organelle movement.
4. Explain the following terms:
 - a. Cytosol
 - b. Signal recognition protein
 - c. Metacentric chromosome
 - d. pinocytosis
 - e. Autophagy
5. Outline the structural and functional differences between the outer and inner mitochondrial membranes.
6. Illustrate a fully labelled diagram of a nucleosome.
7. Using membrane phospholipids as an example, explain the concept of membrane asymmetry and state its functional significance.
8. Outline four functions of membrane glycolipids.
9. Using labelled diagrams, describe the stages of mitotic cell division.
10.
 - a. Briefly describe the structure and function of a peroxisome.
 - b. Outline 2 clinical features of peroxisomal disorders in humans

SECTION B

Answer two questions in this section. Each question carries equal marks (20 marks each)

11. Give a detailed account of the structural organization and function of microfilaments.
12. Discuss the structure and function of the nuclear envelope.
13. Discuss the role of Endoplasmic Reticulum and Golgi complex in protein glycosylation and trafficking.
14. Using relevant examples and illustrations, describe the mechanisms of active transport across cell membrane.



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MAIN CAMPUS

SZL 203: HISTOLOGY AND CYTOLOGY

Date: 15th February, 2018

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer ALL questions in SECTION A and ANY ONE question from SECTION B.



SZL 203: HISTOLOGY AND CYTOLOGY

Date :

Time: 2 HRS

Instructions

1. Answer **ALL** questions in section A (50 marks) and **ANY ONE** in section B (20 marks)
2. Illustrate your answer(s) with labelled diagrams where appropriate

SECTION A (50marks). Each question carry equal marks (5marks)

- Q1. Using illustrations, describe the four types of epithelial cells
- Q2. Outline the components of a basement membrane
- Q3. Describe the two types of adipose tissue and their significance
- Q4. Write short notes on three (3) types of stains used in Histological processes
- Q5. Distinguish between dense and loose fibrous tissue in cytology
- Q6. Briefly discuss the process of tissue histological procedure
- Q7. Briefly define the following terminologies in regards to muscle fibres (a) Sarcolemma (b) Sarcoplasm (c) Satellite cells
- Q8. Briefly discuss the muscle contraction process
- Q9. Using their structures, briefly distinguish the five (5) types of white blood cells
- Q10. Distinguish between the following three types of neurons based on their functions (a) Motor neurons (b) Sensory neurons (c) Interneurons

SECTION B (20 marks) Answer ANY ONE question in this section

Each question carry equal marks (20 marks)

- Q11. Elastin cartilage is significant in tissue formation and locomotion of cells. Discuss
- Q12. Citing specific examples, discuss the structure of any type of muscles tissue
- Q13. Give a detailed account of the anatomical structure of a nerve cell



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MAIN CAMPUS

SZL 205: ARTHROPOD BIOLOGY

Date: 21st February, 2018

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any THREE from SECTION B
- Use illustrations where appropriate



SZL 205: ARTHROPOD BIOLOGY

Section A. (Answer all questions. Each question carries 4marks)

1. State the morphological and molecular evidence suggesting new arthropod taxonomies as opposed to the traditional approach of classification.
2. State any four functions of biramous appendages in crustaceans.
3. State four unique characteristics of the Class Pycnogonida (the sea spiders)
4. Briefly explain the disadvantages any four strategies used for controlling arthropod pests and disease vectors.
5. With aid of a diagram, distinguish between the capitulum of argacidae and ioxodidae ticks.
6. State four characteristics that make Lepidoptera agriculturally important.
7. Briefly explain the process of ecdysis in a named arthropod.
8. Give a brief account of the life history of ticks.
9. Briefly describe the unique reproductive strategy in Aphids and termites.
10. Briefly explain the social behavior of arthropods.

SECTION B (Answer any three questions, 20marks each)

11. Discuss the characteristics of arthropods that distinguish them from the other animal phyla.
12. Discuss the economic importance of the Class insecta.
13. Discuss the numerous, diverse medical conditions related to arthropods.
14. Advance a critical analysis of the origin and evolution of arthropods.
15. Discuss the habit and life history of tsetse fly.